
Algorithm 3: Relaxed reachability backpropagation

Input:

- V . Set of nodes
- E_s . Set of scheduled edges (v, u, dep, arr)
- E_t . Set of transfer edges (v, w, Δ)
- $\text{country}(v)$. Country associated with node v
- $\text{bucket}(t)$. Time bucket mapping function
- $B_{\max}$. Maximum bucket index within 24-hour horizon

Output:

- $R(v, b)$. Set of reachable countries for all $v \in V, b \in \{0, \dots, B_{\max}\}$

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1: // Initialize reachability sets
2: for all  $v \in V$  do
3:   for  $b = 0$  to  $B_{\max}$  do
4:      $R(v, b) \leftarrow \{\text{country}(v)\}$ 
5: // Build relaxed adjacency lists
6: Initialize  $\text{Adj}(v, b) \leftarrow \emptyset$  for all  $v \in V, b \in \{0, \dots, B_{\max}\}$ 
7: // Scheduled edges
8: for all  $(v, u, dep, arr) \in E_s$  do
9:    $b \leftarrow \text{bucket}(dep)$ 
10:   $b' \leftarrow \text{bucket}(arr)$ 
11:  if  $0 \leq b \leq B_{\max} \wedge 0 \leq b' \leq B_{\max}$  then
12:     $\text{Adj}(v, b) \leftarrow \text{Adj}(v, b) \cup \{(u, b')\}$ 
13: // Transfer edges
14: for all  $(v, w, \Delta) \in E_t$  do
15:   for  $b = 0$  to  $B_{\max}$  do
16:      $t \leftarrow \text{start\_time\_of\_bucket}(b)$ 
17:      $t' \leftarrow t + \Delta$ 
18:      $b' \leftarrow \text{bucket}(t')$ 
19:     if  $0 \leq b' \leq B_{\max}$  then
20:        $\text{Adj}(v, b) \leftarrow \text{Adj}(v, b) \cup \{(w, b')\}$ 
21: // Backward propagation over time buckets
22: for  $b = B_{\max}$  downto 0 do
23:   repeat
24:     changed  $\leftarrow$  false
25:     for all  $v \in V$  do
26:       for all  $(u, b') \in \text{Adj}(v, b)$  do
27:         if  $R(u, b') \setminus R(v, b) \neq \emptyset$  then
28:            $R(v, b) \leftarrow R(v, b) \cup R(u, b')$ 
29:         changed  $\leftarrow$  true
30:   until changed = false
31: return  $R$ 
```

This requires of some preprocessing time, but we only need to run it once and then we have our reachable country sets precomputed to feed our main algorithm at any time we see convenient.